

Research & Project Experience

Hadithya N.W.M.H.
University of Moratuwa
hadithyanwmh.22@uom.lk

In this document, I would like to share my enthusiasm for this opportunity and highlight my key experiences in researching the healthcare domain.

Introduction

As an undergraduate student, I started my journey with the ambition of becoming a researcher, driven by my ability to think critically and analyze different situations. However, along the way, I found myself exploring various technologies, and learning diverse aspects of IT. Looking back, I don't see this as a setback instead, it gave me a strong foundation in my field.

At one point, I reached out to seniors from other departments, hoping to get involved in their research projects, but I didn't get the opportunities I was looking for. So, I decided to take my own approach turning simple ideas into research-driven projects using the knowledge and skills I had. While these projects may not fully align with traditional research, they reflect my passion for solving real-world problems through analytical thinking. Here, I'd like to share two key projects that showcase my efforts in this direction.

Project 1: RhymeFIT (First Year Hardware Project)

GitHub Project: <https://github.com/hhadithya/rhyme-jacket>

Achievements: A Grade for the module

Background

At the beginning of our first-year project, we had to brainstorm different ideas for our proposal. As the team leader, I initially pitched a project on **Dengue Breeding Place Detection using Drones**, which advanced to the second round of evaluations. However, it wasn't feasible at that time due to budget constraints and technical challenges. Interestingly, one of the lecturers, **Professor Gokul [designation may differ]**, was working on a similar project in the mechanical department.

Due to these limitations, I had to shift my focus to a new project. Meanwhile, **Dean sir** also suggested another project related to **soil measurement units**, but my heart was set on working on something impactful for society.

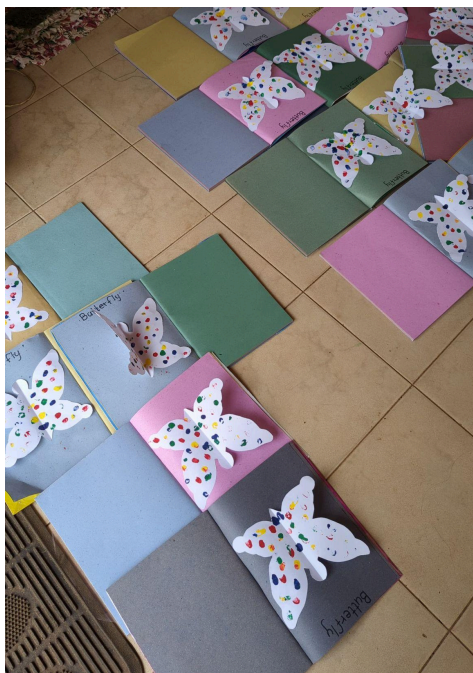
I began researching various disabilities to find a meaningful theme for my project. Initially, I considered developing a **VR headset for disabled individuals** to interact with virtual reality using gestures. Despite having a strong background in VR and AR technologies, I had many doubts about feasibility.

With the guidance of **Dean Sir and Dr. Firdhous**, I shifted my focus to **Autism Spectrum Disorder (ASD)**. I started reading research papers from Google Scholar and ResearchGate to explore existing work and research gaps. Since I also had a background in music, I eventually got the idea to combine music with a health-related issue or personal development for children. That's how **RhymeFIT** was introduced.

Unstructured Research Approach

My idea was to **create a jacket that captures a child's movements and turns them into sound**, which would be integrated into an interactive game. I have always wanted to reason through everything I work on, and that mindset led me to conduct independent research before finalizing the project.

- I visited **Ambasewana Preschool, Piliyandala**, to **observe children's behavior** and discuss with teachers.
- I also visited **Prince of Wales College, Primary Section**, where I had another in-depth discussion about child behavior.

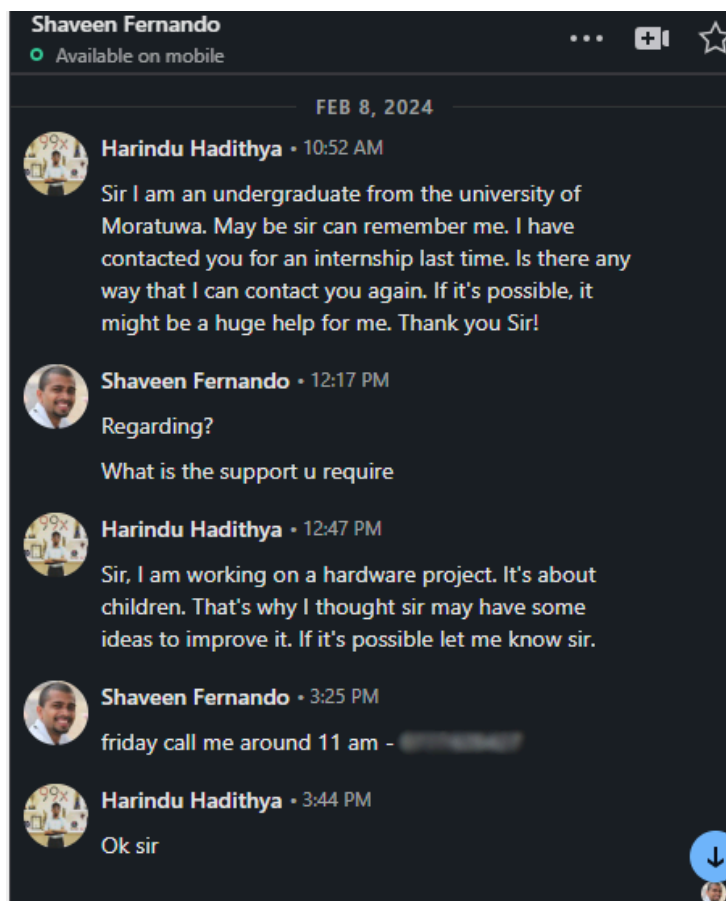


*captured at 08/02/2024 as evidence for my progress
left to right: Ambasewana Preschool Piliyandala ,
Prince of Wales, College*

After analyzing children's capabilities, comfort levels with wearable technology, and technical constraints, I finalized my target sample. While the sample size was small, I ensured the project's feasibility from every aspect based on my available resources.

I didn't stop there. Then I went to different resource persons to get feedback on my project. I searched different people from linkedin to have a chat on these.

- Reached out to professionals via **LinkedIn** and managed to contact someone working with children. After a discussion, he provided valuable **guidance on ethical considerations** when working with children.
- Consulted a **pediatrician at Lanka Hospitals**, who highlighted factors related to **urban child obesity** and how our project could help.
- Discussed feasibility with **Mr. Rukshan J. Senanayaka (19' Batch)**, who advised me to **focus on the happy path first** before tackling more complex scenarios.



Even though I didn't know this resource person directly I tried to contact because of the enthusiasm I had.

Final Result

After handling various dynamic situations, I managed to bring my project to life with the help of my team members. We created a scaled-up version for the demo, but my original aim was

to design a smaller version to test with children. I believe the effort I put in during that time was more valuable than if I had completed everything perfectly. I intended to write a research paper on this project and publish it when the ICITR conference came around, but I missed that opportunity due to other workload demands. I also wanted to pitch this idea in a competition, but I missed that chance as well. Nevertheless, I still believe this project has the potential to evolve into a meaningful research study.



[link for the project post shared on linkedin](#)

Project 2: RemeD

GitHub Project: <https://github.com/HackPack-CS307/RemeD>

Achievements: CodeSprint 7.0 Organized by IIT DesignSprint 1st Place, BuildSprint 5th Place, SprintPitch 10th Place

Background

I worked on this project with my school colleagues before we even entered university. In fact, this project could serve as the key factor for my earlier mentioned project. I applied a similar strategy but expanded on it significantly.

The inspiration for this project comes from an interesting experience. After applying for a competition, we were looking for different ideas. One evening, I needed to pick up some medicine for my sister and found myself going to four different pharmacies in search of the right medication. On my way home, I noticed a small pharmacy nearby that I hadn't considered before, and to my surprise, they had the medication I needed. This experience sparked the idea for my project, and I decided to explore it further.

Unstructured Research Approach

My idea was to connect all pharmacies to our mobile application, allowing patients to search for medications and find the best possible locations to obtain them. This approach considers both the distances to the pharmacies and the availability of the medications.

To gather human feedback, I followed a similar procedure. First, I visited K-Zone Katubedda, where I randomly selected people and asked whether they believed this type of application would make their lives easier. After that, I approached several pharmacies to discuss how connecting with this application could impact their businesses. Some of them were supportive of the idea, while others were not.

Final Remarks

That's how I have worked on different projects with enthusiasm, even though they were just simple situations, with no real need for research. But I always wanted to turn something into a research idea. Back in my first year, I had an idea to conduct research on **babies' cry patterns** to translate them into actual words, allowing us to understand their real needs. I thought about making it my final-year project. Recently, YouTube suggested a video from Google about this exact project [[link](#)]. That made me reflect, why do I keep giving up on my ideas like this? Then I realized the reason: a lack of understanding of how to conduct proper research. That's why I see this opportunity as a turning point in my life. Even if I'm not selected, I feel like just writing this experience down has helped me take the first step. And honestly, I'm truly proud of the things I've done so far.